

rate. This happens in situations where miners expect good things of the asset in the future, and therefore proactively connect machines to help secure the network. This instills confidence, and perhaps the expected good news has also traveled to the market, so the price starts going up.

Once it's been ascertained that the hash rate is growing, often the best way to compare the relative security of cryptoassets is through a calculation of the equipment securing the network. Using a dollar value is helpful because it gives us an idea of how much a bad actor would have to spend to re-create the network, which is what the actor would need to launch a 51 percent attack.

As of March 2017, a Bitcoin mining machine that produced 14 terahash per second (TH/s) could be bought for \$2,300. The idea of TH/s can be thought of as similar to a personal computer's clock speed, which is often measured in gigahertz (GHz), and similarly represents the number of times a machine can execute instructions per second. It would take 286,000 of the aforementioned 14 TH/s machines to produce 4,000,000 TH/s, which was the hash rate of the Bitcoin network at the time. Hence, Bitcoin's network could be re-created with a \$660 million spend, which would give an attacker control of 50 percent of the network. Yes, 50 percent, because if the hash rate started at 100, and an attacker bought enough to re-create it (100), then the hash rate would double to 200, at which point the attacker has a 50 percent share.

Ethereum's mining network, on the other hand, is less built out because it's a younger ecosystem that stores less value. As